AKASH INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Artificial Intelligence and Data Science

MINI PROJECT SYNOPSIS

AI-BASED COLLEGE FAQ CHATBOT

1. INTRODUCTION

The AI-Based College FAQ Chatbot is an intelligent web-based system designed to assist students in obtaining important college-related information through a simple conversational interface. Students frequently have queries regarding academics, examinations, fees, scholarships, departments, faculty, library facilities, placements, hostel facilities, college timings, and other student services.

Currently, students may need to search through different notices, documents, websites, or contact college offices to obtain such information. The proposed system provides a centralized platform where students can enter their queries in natural language and receive relevant information quickly. The chatbot uses Artificial Intelligence, Natural Language Processing (NLP), and semantic information retrieval to understand student queries and identify relevant information from a college-specific knowledge base.

2. PROBLEM STATEMENT

Students often face difficulty finding accurate and relevant information about various college activities and services. Information may be distributed across different documents, notices, departments, and web pages. A conventional FAQ system may also require students to search manually or use exact keywords. Therefore, there is a need for an intelligent chatbot that can understand natural-language questions and provide relevant college-specific information through a single interface.

3. OBJECTIVES

- To develop an AI-based chatbot for answering frequently asked student queries.
- To provide a simple and user-friendly conversational interface.
- To use Natural Language Processing to understand student questions.
- To retrieve relevant answers from a college-specific FAQ and document knowledge base.
- To support different ways of asking the same question using semantic similarity.
- To provide reliable responses based on approved college information.
- To handle unknown or unsupported questions appropriately.
- To reduce repetitive queries directed towards college administrative departments.
- To provide students with quick and convenient access to college information.

4. PROPOSED SYSTEM

The proposed system is a web-based AI College FAQ Chatbot. Students can enter questions such as “Where is the examination section?”, “How can I apply for a scholarship?”, “What is the attendance requirement?”, “Where is the college library?”, or “How can I contact the placement department?”. The chatbot processes the student's question using NLP techniques and searches the college knowledge base for relevant information. The most relevant information is then returned as a conversational response. The knowledge base can contain verified FAQs, college notices, academic information, student guidelines, and approved PDF documents.

5. METHODOLOGY

Data Collection: Collect and organize verified college FAQs, academic information, examination information, student services information, and relevant approved college documents.

Knowledge Base Creation: Convert collected information into a structured FAQ dataset and process approved college documents such as PDFs and notices.

Natural Language Processing: Process student queries and identify important terms and semantic relationships.

Semantic Search: Compare the student's question with knowledge-base content using vector representations and similarity techniques.

Response Generation: Provide a clear conversational response using the retrieved information and, where appropriate, identify the relevant source document.

Unknown Query Handling: If sufficiently relevant information is unavailable, inform the student rather than providing an unsupported answer.

Testing: Test different question wordings, spelling variations, categories, and unsupported questions to evaluate relevance and reliability.

6. SYSTEM ARCHITECTURE

Student → Web-Based Chat Interface → Flask Backend → NLP / Query Processing → Semantic Search → College FAQ & Document Knowledge Base → Relevant Information → AI Chatbot Response → Student

7. TECHNOLOGIES USED

Component	Technology
Programming Language	Python
Frontend	HTML, CSS, JavaScript
Backend	Flask
Artificial Intelligence	NLP and Semantic Search
Machine Learning	Text similarity / embeddings
Database	SQLite
Knowledge Base	College FAQs and approved documents
Development Environment	Visual Studio Code
Operating System	Windows

8. EXPECTED OUTCOME

The proposed project is expected to produce a functional web-based AI chatbot capable of answering common student queries related to the college. The system will provide quick access to college information, understand natural-language questions, retrieve relevant information from the knowledge base, reduce manual searching, handle unsupported questions safely, and provide a convenient conversational interface.

9. SCOPE OF THE PROJECT

The initial system will focus on commonly required college information such as academics, examinations, fees, scholarships, departments, library, placements, student services, and facilities. The system can be enhanced in the future with voice-based interaction, multilingual support, a mobile application, an administrator dashboard, real-time college notifications, student authentication, integration with college management systems, analytics, advanced retrieval-augmented generation (RAG), and additional institutional documents.

10. HARDWARE REQUIREMENTS

- Windows laptop/desktop.
- Minimum 4 GB RAM; 8 GB RAM or higher recommended.
- Modern processor.
- Internet connection.
- Standard keyboard and mouse.

11. SOFTWARE REQUIREMENTS

- Windows Operating System.
- Python 3.x.
- Visual Studio Code.
- Required Python NLP and machine-learning libraries.
- SQLite database.
- Modern web browser such as Google Chrome or Microsoft Edge.

12. CONCLUSION

The AI-Based College FAQ Chatbot aims to provide an efficient and convenient method for students to obtain college-related information. By combining Artificial Intelligence, Natural Language Processing, semantic search, and a college-specific knowledge base, the system can understand student queries and provide relevant information through a conversational interface. The project demonstrates a practical application of Artificial Intelligence and Data Science in an educational environment and can be further expanded into a comprehensive student-support system.

13. TEAM DETAILS

Project Title	AI-Based College FAQ Chatbot
Department	Artificial Intelligence and Data Science
Semester	5th Semester
Institute	Akash Institute of Engineering and Technology
Student 1	ALAN PS — 1SQ24AD002
Student 2	DHEERAJ PRAMOD — 1SQ24AD008
Student 3	MADHAV AJ — 1SQ24AD016
Student 4	JNANESH K — 1SQ25AD400
Project Guide	_____
Academic Year	2026–2027

14. REFERENCES

- Python Documentation – https://docs.python.org/
- Flask Documentation – https://flask.palletsprojects.com/
- Scikit-learn Documentation – https://scikit-learn.org/
- General references on Natural Language Processing, semantic search, and information retrieval.